BRUFACE
MA1 Mechanical Engineering

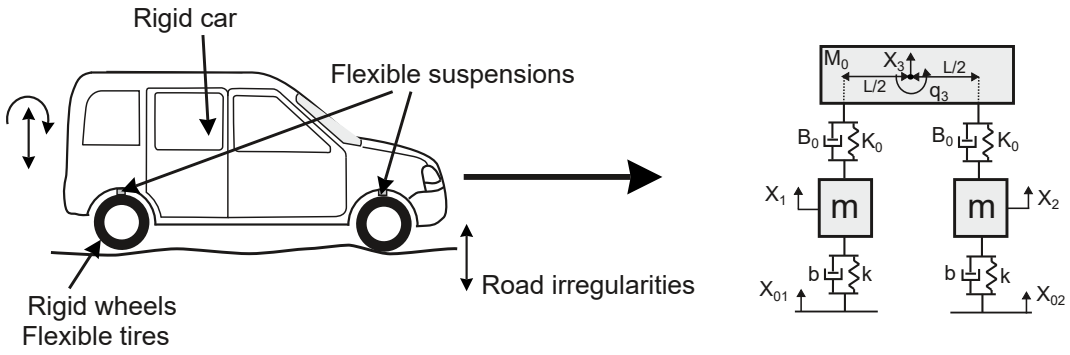
Mechanical Vibrations
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4 DOFS model of a car under road excitation

1 Model and equations

Consider the 4 DOFs model of a car represented below. X_{01} and X_{02} represent the motion imposed on each wheel due to the road profile. The car body is assumed to be rigid and the motion is restricted to a plane, resulting in 2 DOFs (translation, rotation), while each wheel is also assumed to be a rigid body and can only move vertically. K_0, B_0 are the mechanical parameters of the car suspension, while k, b are the stiffness and damping of the tires.



1 Based on the equations of motion derived for each body separately, derive the full set of equations of motion for the car in a matrix form, and determine the mass, damping and stiffness matrices. Use absolute displacements in the equations. Give the expression of the force vector.

2 Assume harmonic excitation and write the equations of motion in the frequency domain. What assumption could you make to have the force vector independent of the frequency? Give the expression of the force vector in this simplified case.

2 Frequency domain computations

Consider the following values of the parameters: M_0=1500 kg, K_0=65 000 N/m, B_0=65 $N/(m/s)$ m=10 kg, k=100 000 N/m, b=10 $N/(m/s)$; $L = 4\ m$, and I_G= 5700 $kg\,m^2$.

2.1 Computation of transmissibilities

1 Compute the mode shapes and the natural frequencies of the car. Give a physical interpretation of each of the mode shapes (what kind of motion of the car is related to each mode shape?)

2 Assuming $X_{01} = X_{02}$, compute and represent the transmissibility X_3/X_{01} using the full model with the coupled equations (no simplification of the force vector). Identify to which mode shapes and resonance frequencies the peaks in the transmissibility curve belong. Which kind of imposed motion is applied when $X_{01} = X_{02}$? Can you make a link with the peaks present in the transmissibility curve?

3 Now, use the simplified force vector derived in point 1.2, compute and represent the transmissibility X_3/X_{01} using the full model with the coupled equations. Compare with the previous case by plotting the two curves on the same graph and comment. Is the simplification used for the force vector valid in your opinion?

4 Assuming $X_{01} = -X_{02}$, compute and represent the transmissibility q_3/X_{01} using the full model with the coupled equations (no simplification of the force vector). Identify to which mode shapes and resonance frequencies the peaks in the transmissibility curve belong. Which kind of imposed motion is applied when $X_{01} = -X_{02}$? Can you make a link with the peaks present in the transmissibility curve?

5 Use the simplified force vector derived in point 1.1, compute and represent the transmissibility q_3/X_{01} using the full model with the coupled equations. Compare with the previous case by plotting the two curves on the same graph and comment. Is the simplification used for the force vector valid in your opinion?

6 The transmissibilities are used in the theory of vibration isolation. Comment on the shape of these and make a link with the theory in the course. In your opinion, which frequency bands correspond to the amplification and the isolation domain? What role do the wheels play in the isolation?

2.2 Projection in the modal basis

1 Using the simplified force vector, project the equations of motion in the modal domain. Are the equations uncoupled? Why?

2 If they are coupled, propose a simplification to decouple the equations of motion, and check its validity by comparing the transmissibility curves X_3/X_{01} (assume $X_{01} = X_{02}$ as above), and q_3/X_{01} (assume $X_{01} = -X_{02}$ as above) obtained with the full model, and with the decoupled modal projection approach. Is your assumption valid?

Time domain computations

Assume now a given road profile w and $X_{01} \neq X_{02}$. Using the equations with absolute coordinates, and using the simplified force vector used above:

1 Knowing that the frequency of excitation depends on the velocity of the car v and the road profile (in particular the amplitude of its different wavelengths) and using a Fourier transform of the road profile, plot the frequency content of the excitation for three different speeds of the car (v=5 m/s, v= 10 m/s and v=30m/s).

2 Compute and represent the absolute displacement $X_3(t)$ for the three car speeds. Use a convolution between the force vector and the impulse responses, and work with the uncoupled equations in the modal basis. (hint: consider X_{01} and X_{02} as two independent excitation sources, and sum the responses using the property of linearity of the system). Use a relative time (t/tmax) in order to plot the curves on the same graph and compare.

3 Represent the frequency content of the three time domain signals obtained. Compare it to the frequency content of the excitation and comment. What differences do you see for the three different velocities and why ?

4 Do you notice a difference in the time domain responses in terms of amplitude? Why ? Can you make a link with the transmissibility curves computed previously? Can you make a link with the theory covered in the chapter about vibration isolation? At which speed(s) is the car body best isolated?

5 Repeat points (2) to (4) for $q_3(t)$ at the three different speeds.